Surya Entrance Test [SET] 2027 - Session 01

Final Question Paper

Math-Sec A (20 Questions | +0 / -0)

Q1. Let A={x∈N: x<20,x is divisible by 2} and B={x∈N: x<20,x is divisible by 3} Find n((A∪B)−(A∩B))

(A) 11

(B) 12

(C) 13

(D) 14

Q2. If f(x) =

$$\begin{cases} x+2, & x<1 \\ 2x-k, & x\geq 1 \end{cases}$$

is continuous at x=1, then k=

(A) 0

(B) 1

(C) 2

(D) 3

Q3. If

$$z = \frac{3+4i}{1-2i},$$

then |z| equals

(A) 1

(B) $\sqrt{5}$

(C) $\sqrt{13}$

(D) 5

Q4. A committee of 4 is formed from 5 men and 4 women. If the committee contains at least two women, the number of possible committees is

(A) 75

(B) 80

(C) 90

(D) 95

Q5. The coefficient of x^4 in $(2+x)^6$ is:

(A) 120

(B) 240

(C) 360

(D) 480

Q6. If

$$2+5+8+\cdots+n=152,$$

then n=

(A) 29

(B) 32

(C) 35

(D) 38

Q7. The perpendicular distance of (3,−2) from

$$3x-4y+10=0$$

is

(A) 3

(B) 4

(C) 5

(D) 6

Q8. The eccentricity of the parabola y^2=12x is

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2

Q9. If $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$ then | A $^{-1}$| equals

(A) $\frac{1}{5}$

(B) $\frac{1}{2}$

(C) -$\frac{1}{2}$

(D) $\frac{1}{6}$

Q10. The value of $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ (A is a identity matrix) is

(A) 0

(B) 1

(C) 2

(D) 3

Q11. Two fair dice are thrown simultaneously. Given that the sum is even, the probability that both numbers are prime is

(A) $\frac{1}{6}$

(B) $\frac{2}{9}$

(C) $\frac{1}{3}$

(D) $\frac{4}{9}$

Q12. How many five-digit numbers can be formed using the digits

$$1, 2, 3, 4, 5, 6$$

without repetition such that the number is divisible by 5?

(A) 96

(B) 100

(C) 120

(D) 144

Q13. $\lim_{x \to 0} \frac{\sin 3x}{\tan 2x}$

(A) $\frac{2}{3}$

(B) $\frac{3}{2}$

(C) $\frac{4}{3}$

(D) $\frac{3}{4}$

Q14. If

$$y = x^x,$$

then

$$\left. \frac{dy}{dx} \right|_{x=1}$$

equals

(A) 0

(B) 1

(C) 2

(D) 3

Q15. The maximum value of

$$x(6-x)$$

is

(A) 6

(B) 8

(C) 9

(D) 12

Q16. $\int_0^1 (3x^2+2x+1)\,dx$ equals

(A) 2

(B) 3

(C) 4

(D) 5

Q17. If

$$\vec{a}=(2,-1,3), \quad \vec{b}=(1,4,-2),$$

then

$$\vec{a}\cdot\vec{b}=$$

(A) -8

(B) -6

(C) -4

(D) 4

Q18. The distance between

$$(1, 2, 3)$$

and

$$(4, 6, 3)$$

is

(A) 3

(B) 4

(C) 5

(D) 6

Q19. The solution of

$$\frac{dy}{dx} = 3x^2,$$

given

$$y = 2atx = 0,$$

has the value of y at x=2

(A) 6

(B) 8

(C) 10

(D) 12

Q20. The maximum value of

$$Z = 3x + 2y$$

subject to

$$x \geq 0, y \geq 0, x + y \leq 4$$

is

(A) 8

(B) 10

(C) 12

(D) 14

Math-Sec B (10 Questions | +0 / -0)

Q21. If

$$z = \frac{1 + i\sqrt{3}}{1 - i},$$

find the value of

$$|z|^2.$$

[Numerical Answer Type — enter integer value]

Q22. The sum of the first n terms of an arithmetic progression is 189. Its first term is 3 and common difference is 2. Find n.

[Numerical Answer Type — enter integer value]

Q23. Find the coefficient of x^5 in

$$(1 + 2x)^8$$

[Numerical Answer Type — enter integer value]

Q24. Evaluate

$$\lim_{x\to 0} \frac{\sin 5x}{x}$$

[Numerical Answer Type — enter integer value]

Q25. If

$$y = x^3 - 6x^2 + 9x + 5,$$

find the x-coordinate of the point where the tangent is horizontal. (Enter the smaller value.)

[Numerical Answer Type — enter integer value]

Q26. Evaluate

$$\int_0^2 (2x+3)\,dx.$$

[Numerical Answer Type — enter integer value]

Q27. If

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix},$$

find

$$|\ A\ |$$

.

[Numerical Answer Type — enter integer value]

Q28. If

$$\vec{a} = (2, 3, -1), \quad \vec{b} = (1, -2, 4),$$

find,

$$\vec{a} \cdot \vec{b}$$

[Numerical Answer Type — enter integer value]

Q29. A bag contains 5 red and 3 blue balls. Two balls are drawn without replacement. Find the probability that both are red. Enter the answer as a decimal rounded to three decimal places.

[Numerical Answer Type — enter integer value]

Q30. If

$$\frac{dy}{dx} = 2x,$$

and

$$y = 3$$

when

$$x = 1,$$

find the value of y at x=3.

[Numerical Answer Type — enter integer value]

Phys-Sec A (20 Questions | +0 / -0)

Q31. A particle moves along a straight line. During the first half of its total journey it travels with constant speed v. During the remaining half, it accelerates uniformly from v to 3v. The average speed for the entire journey is

(A) 2v

(B) $\frac{9v}{5}$

(C) $\frac{8v}{5}$

(D) Depends on acceleration

Q32. If the angle is increased slightly [IMAGE] The coefficient of static friction is just enough to prevent slipping. If the angle is increased slightly,

(A) Normal reaction increases

(B) Friction reverses direction

(C) Friction becomes insufficient

(D) Weight decreases

Q33. A person pushes a wall with constant force for 5 minutes. Which statement is correct?

(A) Work is done because force exists.

(B) Work is zero because displacement is zero.

(C) Energy is transferred to the wall.

(D) Momentum of wall changes.

Q34. A stone tied to a string moves in a horizontal circle. [IMAGE] If the string suddenly breaks,

(A) Stone moves radially outward.

(B) Stone falls vertically.

(C) Stone moves tangentially.

(D) Stone continues circular motion.

Q35. A satellite is in a circular orbit of radius 2R around Earth. Its engines are fired briefly so that its speed becomes 25% greater than the circular orbital speed at that point. What happens immediately after the impulse?

(A) It escapes Earth's gravity.

(B) It moves in an elliptical orbit with the firing point as perigee.

(C) It moves in an elliptical orbit with the firing point as apogee.

(D) It continues in the same circular orbit.

Q36. A particle executes SHM with amplitude A. When its displacement is A/2, the ratio of its kinetic energy to potential energy is

(A) 1:3

(B) 3:1

(C) 2:1

(D) 1:2

Q37. Two rods of identical dimensions are made of copper and glass. One end of each is heated equally. Which reaches thermal equilibrium first?

(A) Copper

(B) Glass

(C) Both simultaneously

(D) Cannot be predicted

Q38. One mole of an ideal gas is taken from state A to state C through two different paths. Path I: Isochoric heating followed by isobaric expansion. Path II: Isothermal expansion followed by isochoric heating. Which quantity remains the same for both paths?

(A) Heat supplied

(B) Work Done

(C) Change in internal energy

(D) Entropy change

Q39. A wave travels along a string. Doubling frequency while keeping wave speed constant causes

(A) Wavelength doubles

(B) Wavelength halves

(C) Speed doubles

(D) Amplitude halves

Q40. [IMAGE] Point P is exactly midway. Electric field at P is

(A) Zero

(B) Towards +Q

(C) Towards -Q

(D) Perpendicular to line joining charges

Q41. A capacitor C is charged to V. It is disconnected from the battery and another identical uncharged capacitor is connected in parallel. The loss in stored energy is

(A) $\frac{CV_2^2}{8}$

(B) $\frac{CV_2^2}{4}$

(C) $\frac{CV_2^2}{2}$

(D) Zero

Q42. [IMAGE] Equivalent resistance between A and B is

(A) 1 Ω

(B) 2 Ω

(C) 4 Ω

(D) 6 Ω

Q43. An ideal transformer has turns ratio 1:5. Primary current is 10 A. Secondary current is

(A) 2 A

(B) 10 A

(C) 25 A

(D) 50 A

Q44. [IMAGE] The refracted ray bends toward the normal. This implies

(A) Speed increases

(B) Frequency decreases

(C) Wavelength decreases

(D) Energy increases

Q45. A convex lens forms a real image of an object. If the upper half of the lens is covered, the image formed is

(A) Half the height

(B) Complete image with reduced brightness

(C) No image

(D) Inverted upper half only

Q46. [IMAGE] The graph represents

(A) Planck's radiation law

(B) Einstein photoelectric equation

(C) Wien's law

(D) Rutherford scattering

Q47. Which statement best explains why nuclei remain stable despite proton-proton repulsion?

(A) Gravity dominates.

(B) Strong nuclear force acts over short distance.

(C) Electrons neutralize proton repulsion.

(D) Magnetic force binds nucleons.

Q48. [IMAGE] The diode is forward biased. If temperature increases,

(A) Forward resistance generally decreases.

(B) Forward resistance increases.

(C) Current becomes zero

(D) Battery voltage reverses.

Q49. A logic circuit gives output HIGH only when both inputs are LOW. The gate is

(A) NOR

(B) NAND

(C) XOR

(D) AND

Q50. A conducting rod of mass m and resistance R slides down frictionless rails inclined at angle θ. A uniform magnetic field acts vertically downward. After a long time, the rod moves with terminal velocity. Which statement is correct?A conducting rod of mass m and resistance R slides down frictionless rails inclined at angle θ. A uniform magnetic field acts vertically downward. After a long time, the rod moves with terminal velocity. Which statement is correct?

(A) Mechanical power equals magnetic energy stored.

(B) Gravitational power equals Joule heating.

(C) Induced current becomes zero.

(D) Magnetic force becomes zero.

Phy-Sec B (10 Questions | +0 / -0)

Q51. A particle starts from rest and moves with constant acceleration $4 m/s^2$.After covering 24m, the acceleration is suddenly reversed to $-2 m/s^2$. Determine: The maximum speed attained.

[Numerical Answer Type — enter integer value]

Q52. A river is 120m wide and flows eastward with a speed of 3m/s. A swimmer can swim at 5m/s relative to water. He wishes to reach the point directly opposite his starting position. Find: The angle at which he should swim with respect to the perpendicular to the bank.

[Numerical Answer Type — enter integer value]

Q53. A 5kg block rests on a rough horizontal surface (μ=0.4). A horizontal force of 40N acts on it. Take $g=10 m/s^2$. Find: The frictional force acting on the block.

[Numerical Answer Type — enter integer value]

Q54. A particle of mass 2kg moves along the x-axis under the force $F(x)=20-2x$ where F is in newtons and x is in metres. If it starts from rest at x=0, find its speed when it reaches x=8m.

[Numerical Answer Type — enter integer value]

Q55. A stone tied to a string of length 1m is whirled in a vertical circle. The minimum speed at the highest point required to keep the string taut is maintained. Find: Speed at the highest point + Tension at the lowest point. Take $g=10\ ms^2$

[Numerical Answer Type — enter integer value]

Q56. A satellite moves in a circular orbit at a height equal to Earth's radius above the surface. Take: Earth's radius R=6400km g=10m/s^2 Determine: Orbital speed + Orbital period.

[Numerical Answer Type — enter integer value]

Q57. A solid cylinder of mass 4kg and radius 0.5m rolls without slipping down an incline through a vertical height of 5m. Take g=10m/s^2. Find: Linear speed at the bottom + Angular speed.

[Numerical Answer Type — enter integer value]

Q58. A steel rod of length 2m at 20°C is heated to 220°C. Coefficient of linear expansion

$$\alpha = 1.2 \times 10^{-5}\,{}^\circ\mathrm{C}^{-1}$$

Find: Increase in length - Final length.

[Numerical Answer Type — enter integer value]

Q59. A particle executes SHM with amplitude 0.20m and angular frequency 5rad/s. Determine: Maximum speed + Speed when displacement is 0.12m + Maximum acceleration.

[Numerical Answer Type — enter integer value]

Q60. A stretched string 2.0m long is fixed at both ends. Wave speed on the string is 160m/s. The string vibrates in its 4th harmonic. Find: Wavelength + Frequency - Distance between two consecutive nodes.

[Numerical Answer Type — enter integer value]

Chem-Sec A (20 Questions | +0 / -0)

Q61. The ion X^{2+} has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6$ The element X most likely belongs to

(A) Group 2

(B) Group 13

(C) Group 14

(D) Group 16

Q62. A gaseous mixture contains 2 mol of He, 3 mol of N_2 and 1 mol of CO_2. If the total pressure of the mixture is 900 mmHg, the partial pressure of nitrogen is

(A) 300 mm Hg

(B) 450 mm Hg

(C) 500 mm Hg

(D) 600 mm Hg

Q63. Which statement is always correct?

(A) Electron affinity increases from left to right in every period.

(B) Atomic radius always decreases down a group.

(C) First ionization enthalpy of Be is greater than that of B.

(D) Electronegativity of noble gases is highest.

Q64. Which species has the greatest bond angle?

(A) $\mathrm{NH_3}$

(B) $\mathrm{H_2O}$

(C) $\mathrm{BF_3}$

(D) $\mathrm{CH_3}$

Q65. A gas occupies 6 L at 2 atm and 300 K. Its volume at 1 atm and 450 K will be

(A) 9 L

(B) 12 L

(C) 18 L

(D) 24 L

Q66. For a process, q=+120J Work done by the system = 40J The change in internal energy is

(A) +80 J

(B) +120 J

(C) +180 J

(D) -180 J

Q67. For the equilibrium 2A(g)⇌B(g) Increasing pressure at constant temperature shifts equilibrium

(A) Towards A

(B) Towards B

(C) No Shift

(D) Depends on catalyst

Q68. In acidic medium, $\mathrm{MnO_4^-}$→Mn^{2+}

(A) +5

(B) -5

(C) -7

(D) +7

Q69. The standard reduction potentials are

$$E^\circ_{\mathrm{Cu^{2+}/Cu}} = +0.34\mathrm{V}$$

$$E^\circ_{\mathrm{Zn^{2+}/Zn}} = -0.76\mathrm{V}$$

The standard EMF of the Daniell cell is

(A) 0.42 V

(B) B) 0.76 V

(C) 1.10 V

(D) 1.52 V

Q70. The coordination number of cobalt in

$$[\mathrm{Co(en)_2Cl_2}]^+$$

(where en = ethylenediamine) is

(A) 2

(B) 4

(C) 6

(D) 8

Q71. At equilibrium,

$$\mathrm{PCl_5(g)} \rightleftharpoons \mathrm{PCl_3(g)} + \mathrm{Cl_2(g)}$$

the equilibrium constant K_p increases when the temperature is increased. Which statement must be correct?

(A) The forward reaction is exothermic.

(B) The forward reaction is endothermic.

(C) The reaction is irreversible.

(D) The value of K_p is independent of pressure.

Q72.A galvanic cell is constructed using the half-cells:

$$Ag^+/Ag,\ E^\circ = +0.80\,V$$

$$Fe^{3+}/Fe^{2+},\ E^\circ = +0.77\,V$$

Under standard conditions, the cell EMF is:

(A) 0.03 V

(B) 1.57 V

(C) −0.03 V

(D) 0.80 V

Q73. During electrolysis of aqueous $CUSO_4$ using inert electrodes, 0.20 mol of electrons pass through the cell. The maximum mass of copper deposited is (Atomic mass of Cu = 63.5)

(A) 3.175 g

(B) 6.35 g

(C) 12.70 g

(D) 25.40 g

Q74. For a first-order reaction, the half-life is 40 min. The fraction of reactant remaining after 120 min is

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{1}{8}$

(D) $\frac{1}{16}$

Q75. Which complex is expected to show optical isomerism?

(A) $[PtCl_4]^{2-}$

(B) $[Co(en)_3]^{3+}$

(C) $[Ni(CN)_4]^{2-}$

(D) $[Ag(NH_3)_2]$

Q76. The major product obtained when tert-butyl bromide reacts with aqueous KOH at room temperature is

(A) tert-Butyl alcohol

(B) Isobutene

(C) n-Butyl alcohol

(D) tert-Butyl chloride

Q77. Which compound is most acidic?

- **(A)** Ethanol
- **(B)** Water
- **(C)** Phenol
- **(D)** p-Methoxyphenol

Q78. Which compound gives a positive Tollens' test but does not undergo aldol condensation?

- **(A)** Acetaldehyde
- **(B)** Formaldehyde
- **(C)** Acetone
- **(D)** Benzaldehyde

Q79. Arrange the following in increasing order of basic strength in aqueous solution: I. Aniline II. Ammonia III. Ethylamine

- **(A)** I < II < III
- **(B)** II < I < III
- **(C)** III < II < I
- **(D)** I < III < II

Q80. Which statement is correct?

- **(A)** Glycine has two chiral carbon atoms.
- **(B)** Glucose is a non-reducing sugar.
- **(C)** Proteins are polymers of α-amino acids.
- **(D)** Sucrose gives a positive Tollens' test.

Chem-Sec B (10 Questions | +0 / -0)

Q81. A sample containing 11.2 L of an ideal gas at STP is completely converted into another gaseous product without any change in the number of moles. Find the number of molecules present in the product.

[Numerical Answer Type — enter integer value]

Q82. A gas absorbs 250 J of heat and performs 80 J of work. Find the change in internal energy (J).

[Numerical Answer Type — enter integer value]

Q83. The concentration of H^+ ions in a solution is $1 \times 10^{-4} M$.

Find the pH of the solution.

[Numerical Answer Type — enter integer value]

Q84. Calculate the volume (L) occupied by 2 mol of an ideal gas at STP.

[Numerical Answer Type — enter integer value]

Q85. During electrolysis, 9650 C of charge is passed through a solution of $AgNO_3$.

Find the mass of silver deposited (g). (Atomic mass of Ag = 108)

[Numerical Answer Type — enter integer value]

Q86. A first-order reaction has a half-life of 40 min. Find the time (min) required for 87.5% decomposition.

[Numerical Answer Type — enter integer value]

Q87. Calculate the molarity when 5 mol of solute is dissolved to make 2.5 L of solution.

[Numerical Answer Type — enter integer value]

Q88. Find the oxidation state of Co in

$$[Co(NH_3)_5Cl]Cl_2$$

[Numerical Answer Type — enter integer value]

Q89. The magnetic moment of a transition metal ion is measured as 4.90 BM. Find the number of unpaired electrons.

[Numerical Answer Type — enter integer value]

Q90. For the Daniell cell,

$$Zn|Zn^{2+}||Cu^{2+}|Cu$$

Given: * $E^\circ_{Zn^{2+}/Zn} = -0.76\,V$ * $E^\circ_{Cu^{2+}/Cu} = +0.34\,V$ Find the standard cell potential (V).

[Numerical Answer Type — enter integer value]